

Exploring Factors of Price, Age, and Region of Origin as Predictors of Online Community Wine Ratings

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Introduction

Research Question 1: Are fluctuations in year and price of wine associated with a higher wine rating?

Research Question 2: Does the origin country of wine (Old World vs New World) impact its rating?

Significance of Current Research

- We seek to build upon the previous research on factors affecting wine quality evaluations, through investigating a large dataset of wine ratings which includes various countries
- Older wines have been to shown to be associated with an “increased fruit flavor, lower astringent and reductive flavor” which all lead to a higher wine quality (Echave et al., 2021)
- Wine experts and traders exhibit implicit biases favoring Old World wines (e.g., French and Swiss) over New World wines (e.g., Brazilian and Argentinean), often rating the former more favorably and describing them with more positive sensory language (Chauvin et al., 2024)
- Country of origin influences wine traders’ mental representations, with traditional Old World producers benefiting from strong country images, while New World producers like Argentina face challenges when introducing wines that deviate from expected national styles (Rodrigues et al., 2020)
- Old World consumers rely more heavily on country-of-origin information in shaping their perception and categorization of wines, while New World consumers focus more on the sensory qualities of the wine regardless of its origin (Valentin et al., 2021)
- Research can shed light on factors that influence wine ratings, a key component in consumer decision-making and market value; by identifying how variables like price, age, and region of origin relate to perceived wine quality, the study can offer insights for both wine producers and consumers.
- Hypothesis:** Older, more expensive, and “Old World” wines will have higher odds of being rated more highly

Methods

Source Website

- Gathered data from Vivino.com, world’s largest wine marketplace containing reviews from community members
 - Found subset from Kaggle file containing data points for 33,605 different wine bottles; includes five smaller sub-datasets with variables for each type of wine: Red, White, Rose, Sparkling, and Varieties
- Modeled red/white wine datasets; ($n_r = 8666$, $n_w = 3764$)
- Both datasets include variables name, country, region, winery, numerical rating (1-5 scale), number of ratings, price (in Euros), and year of production

Data Wrangling

- Of 30 countries included in each dataset, 20 countries were classified as Old World Wine Countries (European wines, e.g., France, Italy); other 10 countries were New World Wine Countries (e.g., Australia, United States)
- For binomial logistic analysis, ratings equal to/above median rating were labeled as “high” rating, ratings below median rating were labeled as “low” rating ($med(r) = 3.9$, $med(w) = 3.8$)
- Price was transformed to $\log(\text{Price})$ due to significant right skew of price observations, since a few wines were very expensive (range: €3.6 - €3410.8)
- Year was calculated as years since 2020: year in which data was collected

Model Building

- Logistic regression model $\log P(Y = 1)/1 - P(Y = 1) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$ for both red and white wine datasets
- With our selected variables plugged in, we obtain the model $\text{Rating} = \beta_0 + \beta_1 \text{Age} + \beta_2 \log(\text{Price}) + \beta_3 \text{World}$

Results

- Red wine model: $-7.23733 + 2.60882\text{LogPrice} - 0.09016\text{Age} + 0.21091\text{OldWorld}$ ($p < 0.05$ for all covariates)
 - Odds of wine bottle having a high rating — 1 unit increase in LogPrice = odds increase by 13.58, 1 unit increase in Age = odds decrease by 8.62%, Old World wine = odds 1.23 higher than New World wine
- White wine model: $-6.78458 + 2.81504\text{LogPrice} - 0.26340\text{Age} + 0.95035\text{OldWorld}$ ($p < 0.05$ for all covariates)
 - Odds of wine bottle having a high rating — 1 unit increase in LogPrice = odds increase by 16.69, 1 unit increase in Age = odds decrease by 23.16%, Old World wine = odds 2.59 higher than New World wine
- Addition of interaction terms between $\text{LogPrice} * \text{Age}$ and $\text{LogPrice} * \text{World}$ yields significant results
 - Effect of LogPrice on rating outcome is stronger as Age increases; Effect of LogPrice on rating outcome is weaker for Old World wines

Figures 1 & 2. Logistic Regression Model Predictions ($x = \log(\text{Price})$, $y = \text{predicted probability}$)

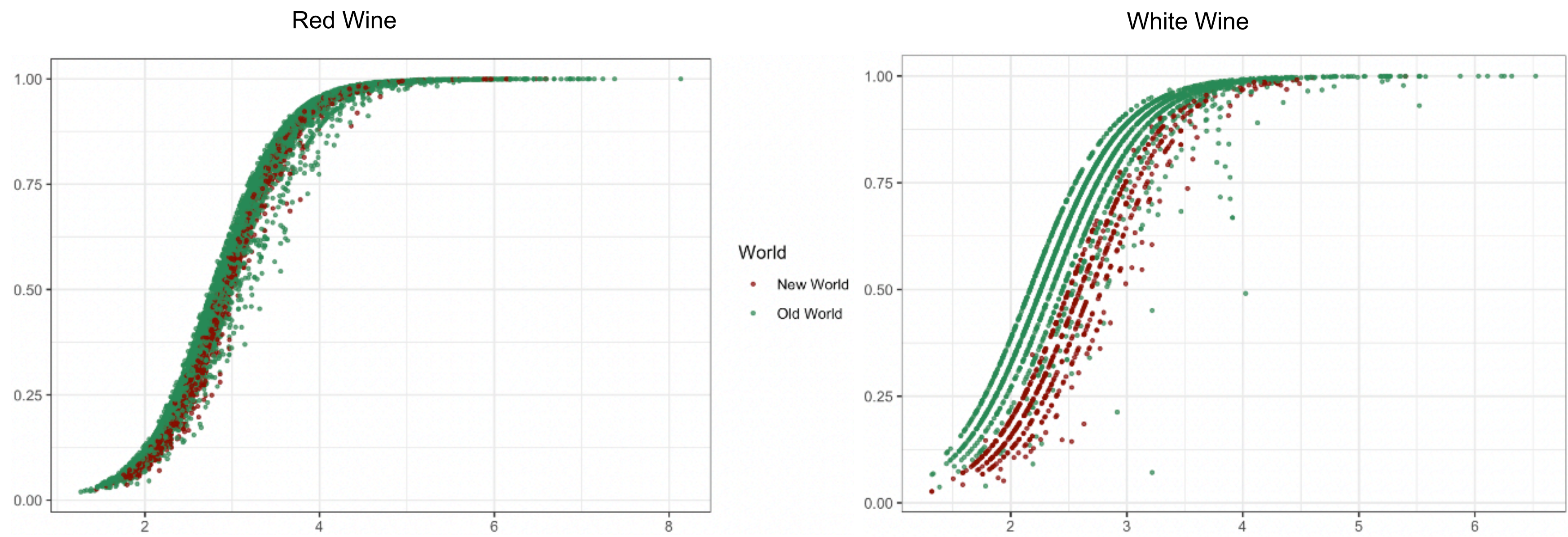


Figure 3. Mean Red Wine Ratings by Country ($n \geq 50$)

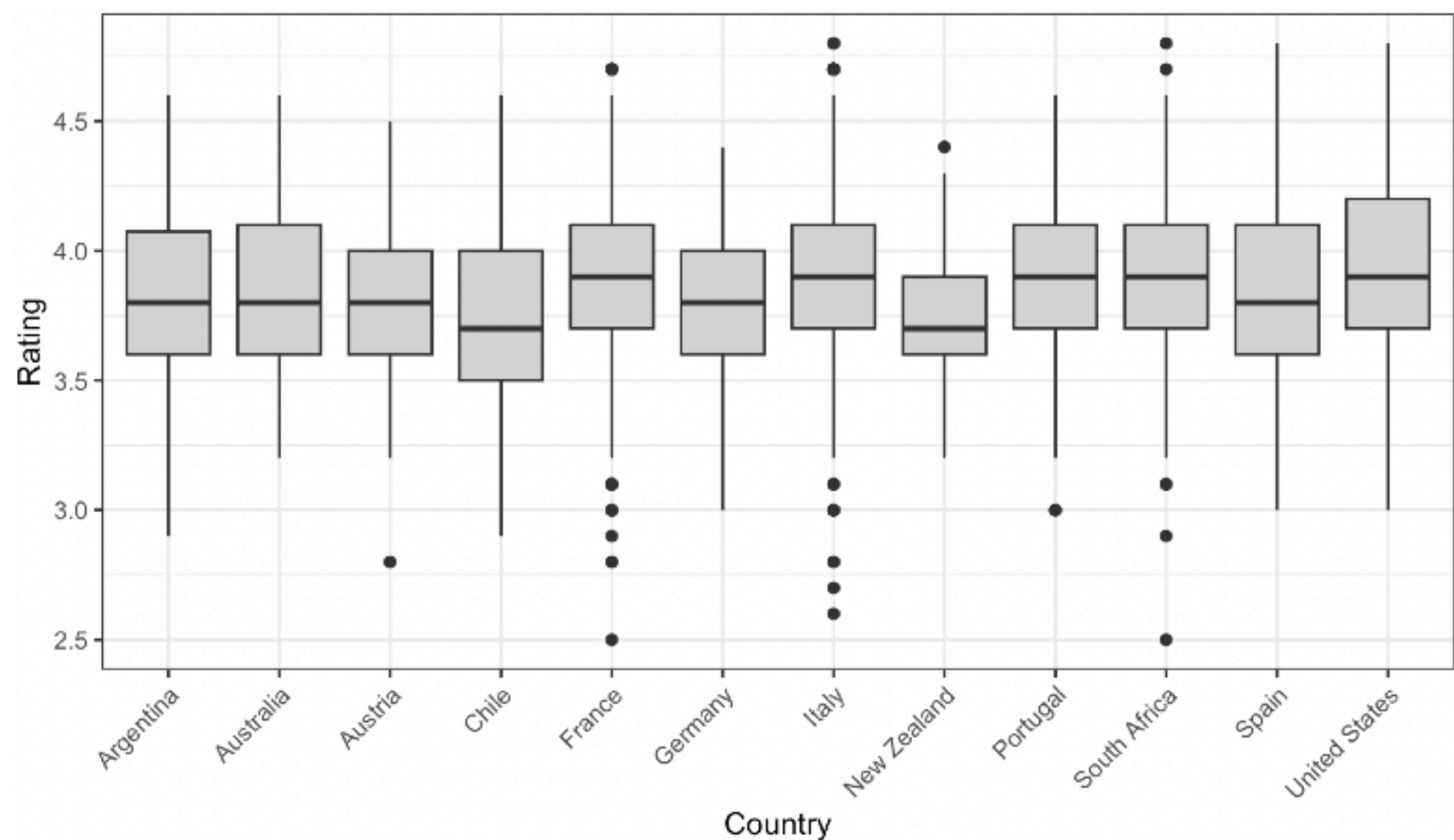


Figure 4. Mean White Wine Ratings by Country ($n \geq 50$)

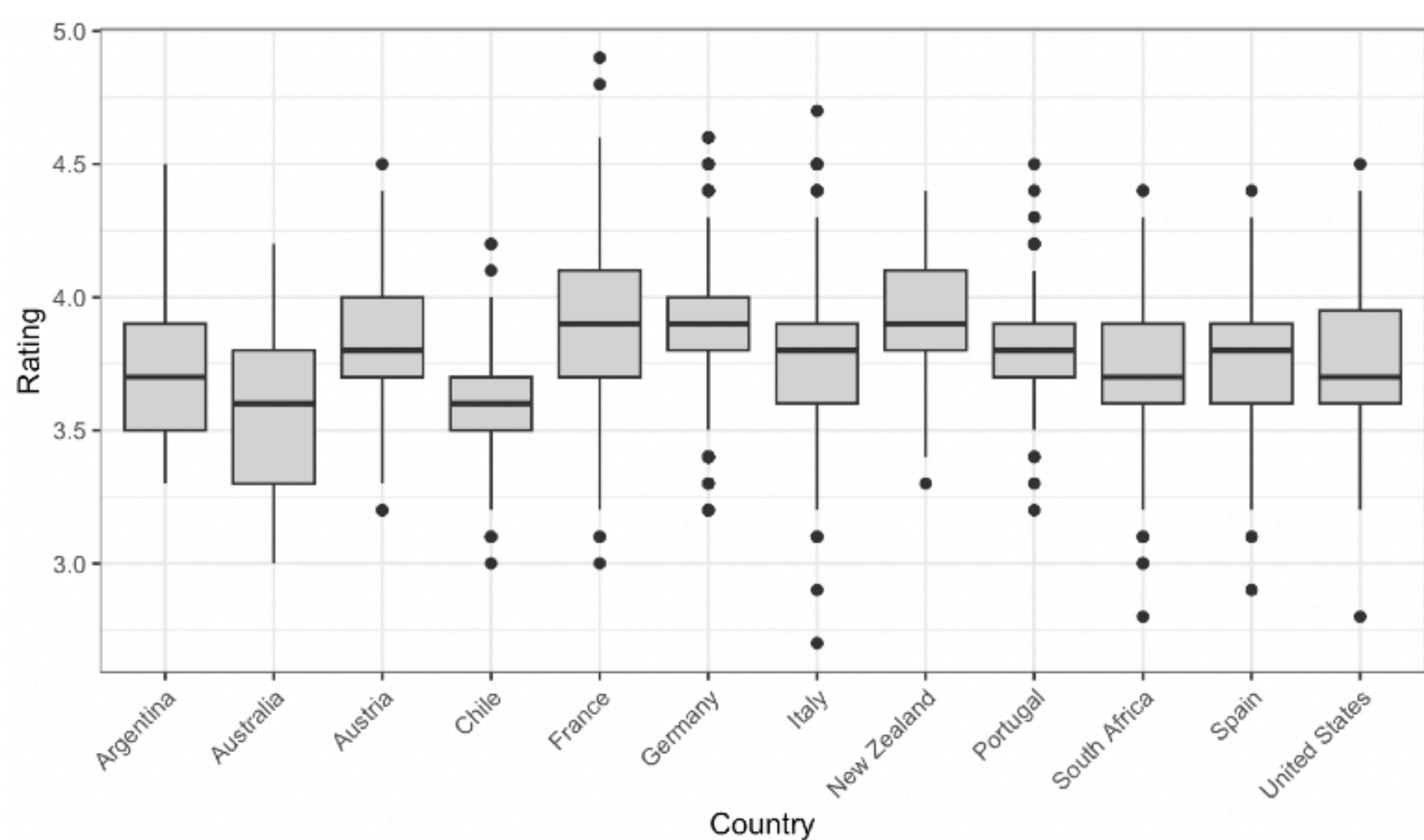


Table 1. Mean Red Wine Ratings by Age Range

Age	Mean Rating	n
1-5	3.84	5945
6-10	3.97	2185
11-15	4.12	405
16-20	4.2	76
21-32	4.28	47

Table 2. Mean White Wine Ratings by Age Range

Age	Mean Rating	n
1-5	3.81	3549
6-10	3.85	182
11-15	4.1	17
16-27	4.11	9

Discussion

Key Takeaways

- Within both the red and white wine models, higher prices, a lower age, and production in an Old World Country were associated with greater odds of having a High rating
- Effects of price, age, and region of origin were stronger in white wine model
- Simpson’s Paradox explained with age variable since price mediates effect of age on Rating; challenges assumptions that higher price and older age equates to higher rating, reveals a more nuanced relationship.

Generalizability

- Website: Wine ratings from one specific website by a unique user base; unable to generalize our results beyond the Vivino website
- Time: Since data was collected 4 years ago, may be differences if the same model were to be fitted with current data (majority of wines in the dataset were produced in the last 4 years); generalizability of our results is thus limited to the short-term of the time of data collection
- Unable to make any causal inferences from the data as there was no manipulation of variables or random assignment procedures with the data

Limitations

- Uncertainty with data collection method from Vivino prevents us from being able to assume random selection and validly interpret results
- Bias: geographic bias due to the nature of Vivino website being most popular in North America and Europe (location of its main offices are in Denmark and the United States); selection bias due to the most popular and available wines being viewed and rated more frequently

Future Research Directions

- Future directions for research could involve the inclusion of more wine types, more total wines, and perhaps deeper dives into the regions within countries and signature wines (e.g., Cabernet Sauvignon vs Pinot Noir)
- Adding more variables (e.g., chemical profiles, winery) would be useful to strengthen the model and examine more detailed differences in wine ratings based on specific attributes of different wines
- Setting baseline point for “High/Low” rating could be adjusted to different percentiles beyond 50th percentile to see if differences emerge

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